SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Swap 2 No's

SWAP 2 NO'S

C Program to Swap Two Numbers.
Swapping two numbers in C programming means swapping the values of two variables.
For example, there are two variables m & n. Value of m is “2” & value of n is “3”.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main()
{
  int a,b,temp;
  scanf("%d%d", &a,&b);
  printf("before swapping a=%d b=%d",a,b);
  temp=b;
  b=a;
  a=temp;
  printf("\nafter swapping a=%d b=%d",a,b);
  return 0;
}
```

INPUT

10
20

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Arithmatic

ARITHMATIC

Write a program that takes two numbers as input and prints their sum, difference, product, and quotient.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  int a,b ;
  printf("enter the two numbers:");
  scanf("%d%d",&a,&b);
  printf("\nsum = %d",a+b);
  printf("\ndifference= %d",a-b);
  printf("\nproduct=%d",a*b);
  printf("\nquotient=%d",a/b);
}
```

INPUT

25
5

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Convert

CONVERT

Write a program that converts Celsius to Fahrenheit. The formula is Fahrenheit = (Celsius * 1.8) + 32.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  float cel,fah;

  printf("enter the temparature in cel");
  scanf("%f",&cel);
  fah = (cel * 1.8)+32;
  printf("\ntemperature in fah = %.2f",fah);
}
```

INPUT

24

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Rectangle

RECTANGLE

Write a program that calculates the area of a rectangle given its length and width. Problem Description: Write a program that calculates the area of a rectangle given its length and width. The program should prompt the user to

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  int l,b ,area;
  printf("enter the value ");
  scanf("%d%d",&l,&b);
  area=l*b;
  printf("area of rectangle=%d",area);


}
```

INPUT

7
8

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Leap

LEAP

Write a program that determines whether a year is a leap year or not. A year is a leap year if it is divisible by 4, but not divisible by 100 unless it is also divisible by 400.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  int a;
  printf("enter the value a");
  scanf("%d",&a);
  if(a%4==0 && a%100!=0 ||a%400==0)
  {
    printf("leap year");
  }
  else
  {
    printf("not a leap year");
  }
}
```

INPUT

2756

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

Max-Min

MAX-MIN

Write a program that takes three integers as input and prints the maximum and minimum of the three.

```c
#include<stdio.h>
int main()
{
  int a,b,c;
scanf("%d%d%d",&a,&b,&c);
if(a >= b && a>=  c)
  printf("max = %d\n",a);
else if (b >= c)
  printf("max = %d\n",b);
else
  printf("max = %d\n",c);
if(a <= b && a<= c)
  printf("min =%d\n",a);
else if (b <= c)
  printf("min = %d\n",b);
else
  printf("min = %d\n",c);
return 0;
}
```

5
6
7

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

ODD EVEN

ODD EVEN

Write a program that checks whether a number is even or odd.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
void main()
{
  int num;
  printf("number is ");
  scanf("%d",&num);
  if(num % 2 == 0)
  {
    printf("even number");
  }
  else
  {
    printf("odd number");
  }
}
```

INPUT

3

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

Sum-Avg

SUM-AVG

Write a program that takes a list of numbers as input and computes the sum and average of the numbers.

```c
#include<stdio.h>
int main()
{
  int a,b,c,d,e, sum;
  float avg;
  scanf("%d%d%d%d%d",&a,&b,&c,&d,&e);
  sum=a+b+c+d+e;
  avg=sum/5.0;
  printf("sum =%d avg= %.2f",sum,avg);
  return 0;
}
```

1
2
3
4
5

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

+ OR -

+ OR -

C program to Check Whether a Number is Positive or Negative.

Problem Description
The program takes the given integer and checks whether the integer is positive or negative.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
void main()
{
  int a;
  scanf("%d",&a);
  if(a>0)
    printf("positive");
  else if(a<0)
    printf("negative");
  else
    printf("zero");
}
```

INPUT

-78

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Greatest

GREATEST

Write a program that takes two numbers as input and determines which one is larger.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  int a,b;
  scanf("%d%d",&a,&b);
  if(a>b)
    printf("a is larger");
  else
    printf("b is larger");
}
```

INPUT

7
9

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Character

CHARACTER

Write a program that takes a character as input and determines whether it is a vowel or a consonant.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<ctype.h>
void main()
{
  char ch1,ch;
  scanf("%c",&ch1);
  ch=tolower(ch1);
    if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u')
    printf("vowel");
  else
    printf("consonant");
}
```

INPUT

A

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Age

AGE

Write a program that takes an age as input and determines whether the person is a child, teenager, adult, or senior citizen.

PROGRAM EDITOR

C

Run

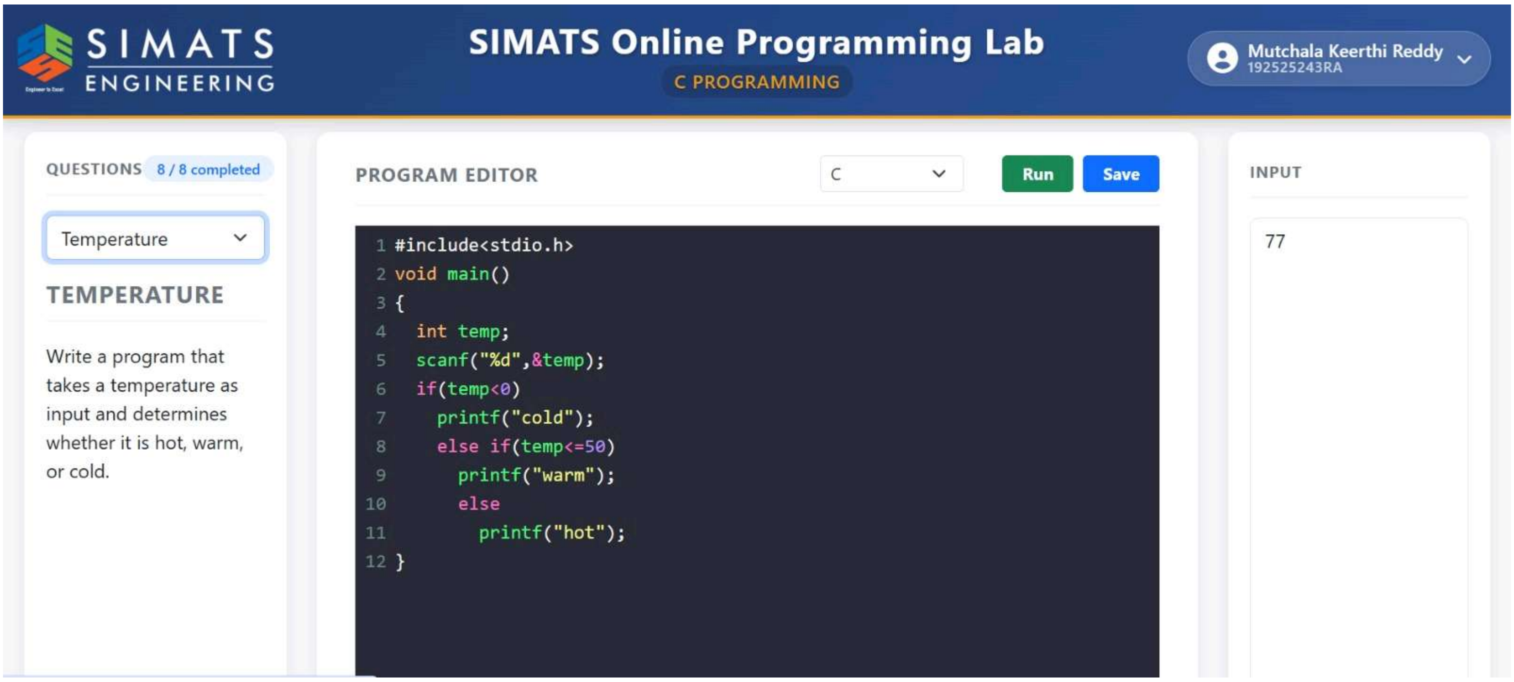
Save

```c
#include<stdio.h>
void main()
{
  int age;
  scanf("%d",&age);
  if(age>13)
    printf("child");
  else if(age>20)
    printf("teenager");
  else if(age>60)
    printf("adult");
    else
     printf("senior citizen");

}
```

INPUT

56


SIMATS ENGINEERING
SIMATS Online Programming Lab
C PROGRAMMING
Mutchala Keerthi Reddy
192525243RA
QUESTIONS
8 / 8 completed
Temperature
TEMPERATURE
Write a program that takes a temperature as input and determines whether it is hot, warm, or cold.
PROGRAM EDITOR
C
Run
Save
#include<stdio.h>
void main()
{
int temp;
scanf("%d",&temp);
if(temp<0)
printf("cold");
else if(temp<=50)
printf("warm");
else
printf("hot");
}
INPUT
77

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Ups-Lows

UPS-LOWS

Write a program that takes a character as input and determines whether it is uppercase or lowercase.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>

void main()
{
  char ch;
  scanf("%c",&ch);

  if(ch >= 'A' && ch <= 'Z')
    printf("upper case");
 else if(ch >= 'a' && ch <= 'z')
   printf("lower case");
  else
    printf("not a alphabet");
}
```

INPUT

r

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Multiple

MULTIPLE

Write a program that takes a number as input and determines whether it is a multiple of 3 or 5.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  int a;
  scanf("%d",&a);
  if(a%3==0)
    printf("multiple of 3");
 else if(a%5==0)
   printf("multiple of 5");
  else
    printf("not a multiple of 3 and 5");
}
```

INPUT

56

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 8 / 8 completed

Grade

GRADE

Write a program that takes a grade as input and determines the corresponding letter grade (A, B, C, D, or F).

PROGRAM EDITOR

C

Run Save

```
# include<stdio.h>
void main()
{
 int a;
 scanf("%d",&a);
 switch(a/10)
 {
   case 9:
     printf("a grade");
     break;
   case 8:
     printf("b grade");
     break;
   case 7:
     printf("c grade");
     break;
```

INPUT

78

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Mutchala Keerthi Reddy
192525243RA

QUESTIONS 1 / 1 completed

Odd or Even

ODD OR EVEN

C Program to Check Whether a Given Number is Even or Odd.

Even Number:
A number is said to be an even number if it is completely divisible by 2.
In other words, if a

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
void main()
{
  int a;
  if(a%2==0)
    printf("even number");
  else
    printf("odd number");
}
```

INPUT

2